

CS301 – IT Solution Architecture: Solution Architecture Final Presentation

(15% of total marks)

1. Instructions

Your team are the solution architects presenting your final architecture design of the solution. **Your audience includes the management, business and technical solution teams.**

1. Keep your presentation to **20 minutes including Q&A**. Presentation slots will be announced separately.
2. As compared to the report, your audience differs. **You need to convince the audience with a clear presentation and a solution that addresses their concerns.**
3. **(Preparation)** If your demo requires startup time, please ensure you get it ready before the presentation. **Pre-recorded videos of your solution for each ASR walkthrough must be done.**
4. **(Presentation)** Explain briefly the business need. The solution addressing the mandatory Features 1-4 specified in the project requirements must be done first before proposing the team's X-factor implementation. Throughout the presentation, please make effort to align your designs, decisions and justifications with the sponsor's requirements, and not just highlight the AWS services used. Be prepared to justify your decisions made. Highlight designs not covered in, or that vary from the course contents. Summarize each member's lessons learnt from the project and course at the end.
5. **(Submission)** Submit your deliverables via **eLearn->Assignment-> G{SectionNo} - Project Submissions** by the specified deadline. **PDF** your proposal deliverable document and name as **G1-{TeamNo}-Project-Final-Presentation.pdf** (e.g., **G1-T1-Project-Final-Presentation.pdf**)
Late submission or failure to comply with the instructions will be penalized.

2. Grading Criteria

1. [5%] Conformance to the instructions.
 - [-] Missing, incomplete or late submission.
2. [5%] Lesson Learnt from the team.
 - [-] Missing participation from most team members
 - [+] Interesting insights
3. [25%] Presentation Conduct
 - [-] Missing team members without valid reasons or missing participation.
 - [-] Poor time management to present and handle Q&A
 - [-] Poorly or not aligning design decisions with the sponsor's requirements
 - [+] Engaging, clear, well-rehearsed.
 - [+] Address questions from sponsors and faculty well
 - [+] Well aligned design decisions with the sponsor's requirements
4. [45%] Solution development for sponsor's functional and non-functional requirements (maintainability, availability, security, and performance.)
 - [-] Missing designs or unable to implement specific sponsor requirements.
 - [-] Wrong conceptual understanding of solution designs.
 - [-] Alternative designs not compared or not well-supported by evidence or reputable references.
 - [+] Bonus features or innovative designs implemented by the team provided basic features are already covered.
 - [+] Quality designs tested with well-supported evidence (e.g., Showing penetration results to compare before and after holds a higher weightage than just mentioning locking down the solution. Showing performance results against baseline holds a higher weightage than just mentioning load balancing is configured.)
 - [+] Quality designs demonstrated by the team. Designs which are explicitly demonstrated hold a higher weightage. (e.g., mentioning availability is handled by AWS service although is counted but holds a lower weightage as compared to demonstrating the design by terminating the instance and availability is not lost).
5. [20%] Solution presented for the team's X-factor.
 - [-] Design is not aligned with the project.
 - [+] Design and implementation demonstrate clear potential impact for the sponsor.

6. Guidelines to the Presentation Contents

As with the video proposal, it is essential to tailor both the structure and content of your presentation to effectively demonstrate that your solution meets the necessary requirements and maintains a high standard of quality.

- **Addressing Sponsor's Requirements and X-factor Requirement**

Following are some guidelines to show your overall solution design.

- Explain the **project context**, e.g., business needs briefly, stakeholders, use cases.
- Explain an **overview of your solution design**, e.g., network, deployment diagrams. **Present your solution in terms of addressing the sponsor's requirements and your team's X-factor requirement. Explain in detail with demonstrations how each requirement is fulfilled.**
- **Overall actual costs** as compared to your planned costs. It can differ.

- **Maintainability Design and Implementation**

Following are some guidelines to show your designs for maintainability.

- Demonstrate how **ease of maintainability** (e.g., design patterns, architectural styles, microservices, multitenancy) are achieved in your proposed architecture. For microservices, explain how the team decides and designs the list of services.
- Explain your **development strategy**, e.g., activity diagram.
- Demonstrate the **development process**, e.g., make a change in the development environment, submit to GitHub, and trigger the necessary testing and delivery/deployment. Show the change on the client.
- Other maintainability designs for better reuse or modifiability in your architecture will be considered.

- **Availability Design and Implementation**

Following are some guidelines to show your designs for availability.

- Demonstrate how the system is still available in the event of a **server software process or hardware or network failures**. For example, when one instance of the Tomcat software process or hardware/laptop hosting the Tomcat software process is not available, another instance can handle the request. At a minimum, one incident of server failure must be allowed. When the network connection between the proxy server and one instance of the hardware/laptop hosting the Tomcat process is not available, the client request is automatically routed to another instance to handle the request.
- Demonstrate how the session data of the user is still available when the failure occurs.
- Other availability designs to eliminate the single points of failure in your architecture will be considered.

- **Security Design and Implementation**

Following are some guidelines to show your designs for security.

- Identify and explain up to **four key vulnerabilities of the architecture**. For each vulnerability, demonstrate with codes or configuration or explanation by words/diagrams how the vulnerability can be addressed during the presentation.
- Ensure the application data are sent over an **encrypted network tunnel**.
- Ensure **proper access control** between the components is configured within the network architecture. E.g., Tomcat instances must only be accessed by the proxy server. The databases should only be accessed by the Tomcat instances.
- Demonstrate how the system is **penetration tested** using the OWASP Zap or a similar tool or an external accreditation site and vulnerabilities are fixed to minimize the attack surface.
- Other security designs to reduce the attack surface in your architecture will be considered.

- **Performance Design and Implementation**

Following are some guidelines to show your designs for performance.

- Demonstrate how to implement a **cache or parallel execution or pre-fetch** can be done to improve the performance of your system. You need to show the baseline and improved performance results.
- Other performance designs to reduce bottleneck in your architecture will be considered.
- Performance demonstrations must follow the workload specifications defined in the implementation report.